

Statistics in Soviet Union

The Law of Large Numbers, Statistical Thought and Data Practice

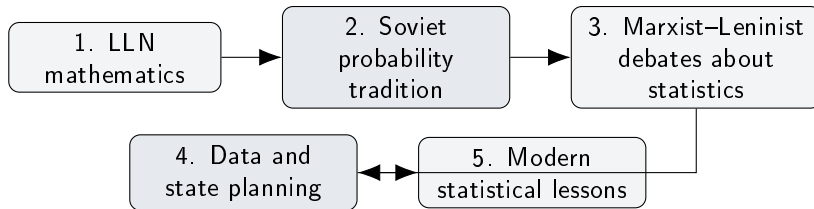
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Scope, Neutrality, and Terminology

- ▶ The subject is the **history of statistical ideas, institutions, and data practices** in the Soviet Union.
- ▶ The purpose is neither to defend nor to criticize communism as a political or economic system.
- ▶ When the slides discuss a “communist view,” they refer specifically to **Soviet Marxist–Leninist doctrine and Soviet institutions**, not to every communist thinker, party, or country.
- ▶ Views also changed across time and differed among Party theorists, state statisticians, economists, and mathematicians.

Source: Mespoulet (2021); Kotz & Seneta (1990); Ghosh (2020)



Central question: What does the LLN guarantee, and what must come from data quality, design, and institutions?

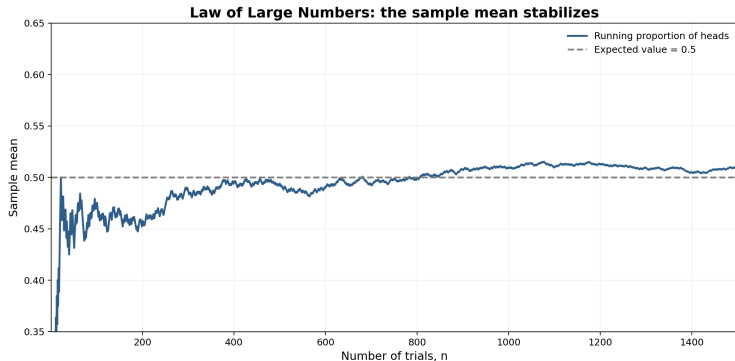
Why Center the Lecture on the LLN?

- ▶ The LLN explains why averages and proportions can stabilize when observations accumulate.
- ▶ It underlies sampling, repeated measurement, Monte Carlo methods, insurance, quality control, and consistency arguments.
- ▶ Soviet history is especially instructive because the USSR produced major LLN mathematics while also debating its role in socioeconomic statistics.

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \longrightarrow \mu$$

The arrow is powerful, but it is conditional on assumptions.

LLN Intuition: Repeated Trials



- Early fluctuations can be large, but the running average tends to stabilize around the expected value.

Weak and Strong Laws of Large Numbers

Let $X_1, X_2, \dots$ be i.i.d. random variables with $E|X_1| < \infty$ and mean μ .

Weak LLN

$$\bar{X}_n \xrightarrow{P} \mu$$

For every $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0.$$

Strong LLN

$$\bar{X}_n \xrightarrow{a.s.} \mu$$

With probability one, the realized sequence of sample means converges to μ .

Source: Standard probability theory; Seneta (2013)

From Chebyshev's Inequality to a Weak LLN

Suppose $X_1, \dots, X_n$ are i.i.d. with mean μ and variance $\sigma^2 < \infty$.

$$E(\bar{X}_n) = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

By Chebyshev's inequality,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

Averaging independent noise reduces variance at the rate $1/n$.

What Makes an LLN Work?

- ▶ A probability model must connect observations to a stable population or data-generating process.
- ▶ Dependence must be limited in a way compatible with the particular LLN being used.
- ▶ The relevant expectation must exist under the theorem's conditions.
- ▶ The observations must actually measure the quantity that the analyst intends to average.

The LLN is a theorem about a probabilistic model—not a guarantee attached to the phrase “large data set.”

What the LLN Does *Not* Guarantee

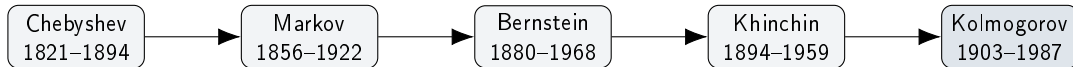
- ▶ It does not remove measurement bias.
- ▶ It does not repair an unrepresentative sampling frame.
- ▶ It does not make inconsistent definitions comparable.
- ▶ It does not prevent strategic or inaccurate reporting.

A useful decomposition

Total error

= random error + systematic error.

The Russian and Soviet Probability Tradition



- ▶ Russian and Soviet mathematicians made major contributions to limit theorems, dependence, stochastic processes, and probability foundations.
- ▶ The LLN therefore had a strong mathematical lineage in the same country where its application to social statistics became controversial.

Source: Seneta (2013); Seneta (2004)

Markov: Large Numbers Can Survive Some Dependence

- ▶ Independence is a convenient condition, but it is not the only route to a law of large numbers.
- ▶ Markov's work helped establish that meaningful limit behavior can arise for dependent sequences.
- ▶ This matters for real data, where observations often interact over time or space.

Statistical implication

The question is not simply “Are the observations independent?” It is “Is the dependence weak or structured enough for averaging to stabilize?”

Source: Seneta (2004); Seneta (2013)

Kolmogorov and the Strong Law

- ▶ Kolmogorov developed major strong-law results in the late 1920s and early 1930s.
- ▶ In the classical i.i.d. setting with $E|X| < \infty$, the sample mean converges almost surely to $E[X]$.

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

This is stronger than saying that the probability of a large deviation becomes small.

Source: Kolmogorov's strong-law work; Seneta (2013)

Kolmogorov's 1933 Axiomatization

Kolmogorov's *Grundbegriffe der Wahrscheinlichkeitsrechnung* placed probability on a measure-theoretic foundation.

$$(\Omega, \mathcal{F}, P)$$

- ▶ Ω is the sample space.
- ▶ $\mathcal{F}$ is a collection of measurable events.
- ▶ P is a probability measure with $P(\Omega) = 1$ and countable additivity.

Historical contrast: abstract probability theory advanced rapidly even while the role of probability in socioeconomic statistics was disputed.

Source: Kolmogorov (1933); Shafer (2018)

Two Meanings of “Statistics” in the Soviet Context

Mathematical probability and statistics

- ▶ Limit theorems
- ▶ Stochastic processes
- ▶ Distribution theory
- ▶ Statistical tests

Socioeconomic and state statistics

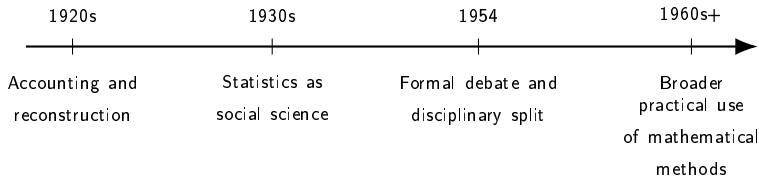
- ▶ Censuses
- ▶ Production reports
- ▶ National balances
- ▶ Administrative accounting

Soviet institutions increasingly treated these as distinct domains, especially from the 1930s through the 1950s.

Source: Mespoulet (2021); Kotz (1965); Ghosh (2020)

What Did the Soviet Communist View Mean?

- ▶ There was **no single timeless view**. The Soviet position changed from the 1920s to the post-Stalin period.
- ▶ Official doctrine and professional practice were not identical: Party theorists, administrators, and mathematicians could disagree.
- ▶ The central doctrinal claim was that socioeconomic statistics should be interpreted through **historical materialism and the social structure being studied**, rather than treated as a universal mathematical science independent of context.



Source: Mespoulet (2021); Ghosh (2020)

Lenin: Statistics, Accounting, and Practical Governance

- ▶ Lenin used large-scale statistical material extensively in political-economic analysis and helped establish Soviet official statistical institutions.
- ▶ In the early Soviet state, he placed strong emphasis on **accounting, counting, and timely figures** for administration and economic reconstruction.
- ▶ He sometimes contrasted inherited nineteenth-century statistical science with a more practical socialist system of accounting and reporting.
- ▶ The emphasis was on statistics as an instrument for **action, control, and planning**, not only as an autonomous academic discipline.

Early Soviet quantification was expected to help govern and build the new economic system.

Source: Kotz & Seneta (1990); Mespoulet (2021)

Stalin-Era Doctrine: Statistics as a Social Science

- ▶ By the 1930s, a dominant official approach described socioeconomic statistics as a **social science** grounded in Marxist–Leninist political economy and historical materialism.
- ▶ Statistics was expected to study mass social and economic phenomena in their historical and class context.
- ▶ Mathematical statistics was increasingly treated as a separate mathematical field, particularly suited to natural and technical sciences.

Doctrinal distinction

Social statistics



Historical and economic interpretation

Separate domain

Probability and mathematical statistics

Source: Rice (1952); Kotz (1965); Ghosh (2020); Mespoulet (2021)

Why “Bourgeois Statistics” Was Criticized

- ▶ Early-1930s Soviet statistical manuals criticized nineteenth-century European statistical theory as being tied to capitalist political economy.
- ▶ The criticism focused especially on interpretations that presented observed social regularities as **stable, natural, or timeless laws**.
- ▶ Soviet Marxist theorists argued instead that social relationships were historically specific and could change when institutions and relations of production changed.

The core disagreement

Does a stable observed frequency describe a **conditional empirical regularity**,
or does it imply a **permanent social law**?

Source: Iastremskii & Khotimskii (1931), discussed in Mespoulet (2021)

The Mean and the “Average Man”

- ▶ Adolphe Quetelet's nineteenth-century social statistics used the idea of an “average man” and stable social rates.
- ▶ Soviet critics interpreted some uses of the mean as encouraging **social fatalism**: the idea that aggregate patterns reflected fixed laws beyond meaningful human intervention.
- ▶ This criticism was about the **interpretation** of averages, not the arithmetic operation of computing a mean.

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

A mean can summarize a population without implying that the underlying social process is permanent.

Source: Mespoulet (2021); Desrosières (1988); Porter (1986)

Why the Law of Large Numbers Became Controversial

- ▶ From the early 1920s, some Soviet Marxist theorists criticized the LLN as a foundation for socioeconomic statistics.
- ▶ Their concern was that stable frequencies could be interpreted as evidence of **inevitable social laws**, reducing the apparent role of deliberate social and political change.
- ▶ The issue therefore concerned the LLN's **philosophical interpretation and domain of application**, especially in social science.

The historical controversy was not simply “Does $\bar{X}_n \rightarrow \mu$?” It was “What does that convergence mean when society itself can change?”

Source: Mespoulet (2021), especially discussion of the mean and LLN

What the LLN Criticism Did—and Did Not—Mean

It often meant

- ▶ Rejecting the LLN as a universal foundation for social statistics.
- ▶ Rejecting interpretations of social frequencies as timeless natural laws.
- ▶ Insisting on historical and economic context.

It did not mean

- ▶ Soviet probability theory stopped developing.
- ▶ The mathematical LLN disappeared.
- ▶ All Soviet scholars held the same philosophical position.

Source: Mespoulet (2021); Seneta (2004); Kotz (1965)

Starovskii's 1936 Attempt at Reconciliation

- ▶ Vladimir Starovskii, a leading Soviet statistician, tried to preserve the usefulness of aggregate statistical laws while making them compatible with Marxist historical interpretation.
- ▶ He did not need to deny that large populations could show regularities.
- ▶ Instead, the key claim was that social regularities were **historically specific**: a law observed under one social system need not remain unchanged under another.

LLN within a regime + distribution may change across regimes

Source: Starovskii (1936), discussed in Mespoulet (2021)

Chance, Planning, and Randomness

- ▶ A planned economy emphasized deliberate organization, targets, and known administrative categories.
- ▶ This created philosophical tension around methods that began from random variation or uncontrolled chance.
- ▶ Soviet statistical practice nevertheless continued to use sampling and probabilistic ideas in selected settings.

The conceptual tension

Planned structure



Random variation

The two are not mathematically incompatible, but Soviet doctrine often required randomness to be interpreted within a prior social or administrative structure.

Source: Mespoulet (2021)

Sampling Under Soviet Statistical Doctrine

- ▶ Russia had an important pre-revolutionary tradition of sample surveys, and sampling continued in the USSR.
- ▶ From the 1930s, Soviet practice often favored **“typical” or stratified designs**: first divide the population into administratively or economically meaningful groups, then select within those groups.
- ▶ Systematic or “mechanical” selection was also widely used.
- ▶ The key difference was often not the calculation itself, but the prior classification of the population and the interpretation of representativeness.

Sampling existed; the dispute was over the role of randomization and probability as the universal justification for social inference.

Source: Mespoulet (2002, 2021); Kruskal & Mosteller (1980)

A Modern Statistical Resolution: Let the Data-Generating Process Change

Instead of assuming one permanent mean μ , let the distribution depend on context t :

$$X_{t,1}, \dots, X_{t,n_t} \sim F_t, \quad E_{F_t}[X] = \mu_t.$$

Within a stable context,

$$\bar{X}_{t,n_t} \longrightarrow \mu_t.$$

Across historical contexts,

μ_t may change.

The LLN describes convergence conditional on a model; it does not assert that a society must have an unchanging distribution.

The 1954 All-Union Conference on Statistics

- ▶ In March 1954, the Academy of Sciences, the Central Statistical Administration, and the Ministry of Higher Education convened a major conference on the foundations of statistics.
- ▶ The conference reaffirmed a Soviet disciplinary distinction: **statistics** was primarily a social science concerned with socioeconomic phenomena, while **mathematical statistics** belonged to mathematics and probability.
- ▶ At the same time, participants argued over how far mathematical and probabilistic methods could and should be used in social analysis.

The result was not a simple rejection of mathematics; it was a boundary drawn around what counted as the foundation of social statistics.

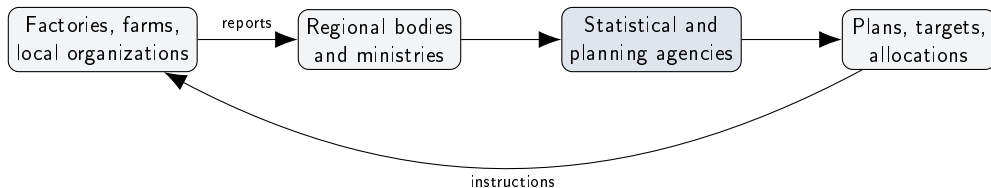
Source: Ostrovitianov (1955); Kotz (1965); Ghosh (2020)

Kolmogorov's Position in the 1954 Debate

- ▶ Kolmogorov defended the scientific importance of probability and the LLN while also warning against unjustified mathematical modeling of social phenomena.
- ▶ This position separated two questions:
 - ① Is the mathematical theorem valid and useful? **Yes.**
 - ② Is every social data set appropriately modeled by a stationary probability law? **Not automatically.**
- ▶ That distinction is very close to modern statistical practice: the theorem may be correct while the model may be inappropriate.

Source: Accounts of the 1954 conference; Kotz (1965); Sheynin's translations of conference materials

Central Planning Was Also a Data Problem



- ▶ Material-balance planning required large flows of information about production, inputs, inventories, labor, and needs.
- ▶ This encouraged extensive administrative reporting and complete enumeration where feasible.
- ▶ The statistical challenge was not merely obtaining many numbers, but obtaining timely, comparable, and accurate numbers.

Source: Mespoulet (2021); IMF et al., *A Study of the Soviet Economy* (1991)

Case Study: The 1937 Soviet Census

- ▶ The January 1937 census produced a preliminary population total of about **162 million**.
- ▶ The result was substantially below official expectations and projections.
- ▶ The census was declared invalid and its results suppressed; senior statistical officials were arrested, and several were later executed.
- ▶ A replacement census was conducted in 1939.

Reported totals

1926 census	147.0 m
1937 prelim.	162.0 m

Source: Merridale (1996); Mespoulet (2021)

The Census as an LLN Lesson

- ▶ A census has extremely large n , so conventional sampling error may be small or absent.
- ▶ Yet a census can still face coverage error, classification problems, processing error, nonresponse, and political or institutional pressures.
- ▶ The 1937 case also illustrates what happens when observed data conflict with strong prior expectations.

The LLN addresses random sampling fluctuation. It does not guarantee correct measurement, processing, interpretation, or acceptance.

Source: Merridale (1996); Mespoulet (2021)

Enterprise Reporting: When Incentives Affect Measurement

- ▶ Enterprises and administrative units reported production and other indicators upward through the planning system.
- ▶ Historical studies document cases in which targets and evaluation rules created incentives that affected how quantities were defined, timed, or reported.
- ▶ The central planning authorities also had incentives to detect inaccurate reporting because plans depended on usable information.

Statistical formulation

Observed data may be written as

$$Y_i = X_i + B_i + \varepsilon_i,$$

where X_i is the target quantity, B_i is systematic reporting bias, and ε_i is random error.

Source: Grossman (1960); IMF et al. (1991); historical studies of Soviet enterprise reporting

The LLN Cannot Average Away Systematic Bias

Suppose

$$Y_i = X_i + B_i, \quad E[X_i] = \mu_X, \quad E[B_i] = \mu_B.$$

Then

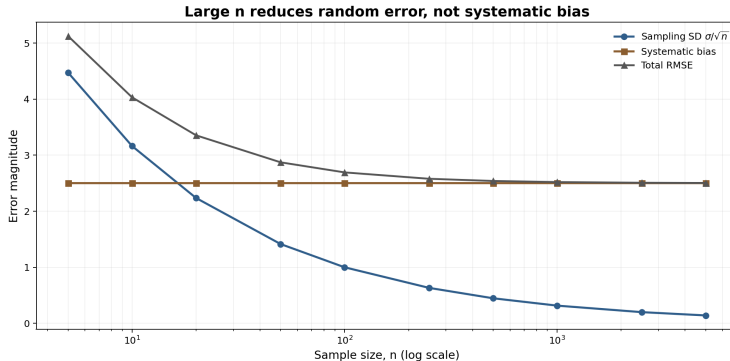
$$\bar{Y}_n = \bar{X}_n + \bar{B}_n.$$

If LLNs apply to both terms,

$$\bar{Y}_n \longrightarrow \mu_X + \mu_B.$$

If $\mu_B \neq 0$, more observations can make the estimate more stable around the *wrong target*.

Large n : Random Error Shrinks, Bias Can Remain



- The LLN is powerful against random fluctuation; systematic error requires better measurement, design, auditing, or modeling.

After Stalin: Change Without a Complete Break

- ▶ The statistical administration regained a more independent institutional role after the late Stalin period, and social statistics expanded again.
- ▶ From the 1950s and especially the 1960s, Soviet practice increasingly used many of the same technical tools found internationally, including sampling and mathematical methods.
- ▶ The ideological distinction between social statistics and mathematical statistics did not disappear immediately, but practical boundaries became less restrictive.
- ▶ Soviet researchers also expanded work in cybernetics, optimization, computing, and mathematical economics.

The Soviet approach to statistics evolved substantially over seven decades; it should not be described as one fixed doctrine.

Source: Mespoulet (2021); Boldyrev (2020); Ghosh (2020)

Takeaways and Selected Sources

Statistical takeaways

- 1 The LLN is a theorem about convergence **under assumptions**.
- 2 A stable aggregate can coexist with a changing data-generating process.
- 3 Large n reduces random error; it does not eliminate systematic bias.
- 4 Data institutions and incentives affect what an average converges to.

Selected historical sources

- ▶ Ghosh, A. (2020). *Making It Count*. Princeton UP.
- ▶ Kotz, S. (1965). "Statistical Terminology—Russian vs. English."
- ▶ Kotz, S. & Seneta, E. (1990). "Lenin as a Statistician." JRSS A.
- ▶ Mespoulet, M. (2021). "Creating a Socialist Society and Quantification in the USSR."
- ▶ Merridale, C. (1996). "The 1937 Census and the Limits of Stalinist Rule."
- ▶ Seneta, E. (2013). "A Tricentenary History of the Law of Large Numbers."

The LLN explains why averages stabilize; data quality determines what they stabilize to.